How-To Library

Informatica

Install Python for the Python transformation on Hadoop

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Abstract

Follow the steps to install Python on each Data Integration Service machine so that the Spark engine can run the Python transformation. This article uses Python 3.6.5 on Cloudera CDH 6.1, but you can follow similar steps for other Python and Hadoop distributions such as Amazon EMR.

Supported Versions

- Informatica Data Engineering 10.4.1.1

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Overview

To run the Python transformation, the Spark engine requires a Python installation. Follow the steps in this article to install Python on each Data Integration Service machine and integrate Python with the Informatica domain.

Environment

This article uses the following environment, but you can follow similar steps for other Python versions and Hadoop distributions such as Amazon EMR.

Informatica Domain	10.4.1.1
Hadoop Cluster	Cloudera CDH 6.1
Python Version	3.6.5

Steps

This article uses the following method to install Python on each Data Integration Service machine:

1. Install Python in a directory that we will set to an environment variable called `$PYTHONHOME`.
2. Install Python third-party libraries in the same location.
3. Copy the contents under `$PYTHONHOME` to `$INFA_HOME/services/shared/spark/python/` where `$INFA_HOME` is the location of the Informatica domain.
4. Repeat steps 1 to 3 for each Data Integration Service machine.

The final location `$INFA_HOME/services/shared/spark/python/` is the location that the Data Integration Service uses to push the Python installation to the Hadoop cluster nodes where the Spark engine runs the Python transformation.

Install prerequisite packages

On the Data Integration Service machine, install the following packages:

OpenSSL

Run the following command to install OpenSSL:

```
sudo yum install openssl-develop
```

zlib

Run the following command to install zlib:

```
sudo yum install zlib-devel
```

libffi

Run the following command to install libffi:

```
sudo yum install libffi-devel
```

Download the Python distribution

Run the following command to navigate to the home directory:

```
cd ~
```

Download the Python distribution onto the Data Integration Service machine. For example, you can run the following command to get an installation of Python 3.6.5 from the internet:

```
wget https://www.python.org/ftp/python/3.6.5/Python-3.6.5.tgz
```

The command downloads the Python distribution in the home directory.

Run the following command to untar the file:

```
tar -xvf Python-3.6.5.tgz
```

The extracted files appear in the directory `~/Python-3.6.5`.

Prepare a directory to install Python

Run the following command to create a `Python3` directory under the home directory:

```
mkdir Python3
```

You can run the command `ls Python3` to see that the `Python3` directory is currently empty.

Run the following command to create an environment variable called `$workingdir` and set the value to the present working directory:

```
export workingdir=`pwd`
```

Run the following command to verify the `$workingdir` environment variable:

```
echo $workingdir
```

Install Python

Run the following command to change directories to the `~/Python-3.6.5` directory where we extracted the Python distribution:

```
cd Python-3.6.5
```

Run the following command to compile Python under `$workingdir/Python3`:

```
./configure --enable-shared --prefix=$workingdir/Python3
```

Run the following commands to build the Python installation:

```
make
make install
```

Set environment variables

Set the following environment variables:

PYTHONHOME

Run the following command to set the `$PYTHONHOME` environment variable:

```
export PYTHONHOME=$workingdir/Python3
```

Notice that `$PYTHONHOME` is the directory where we installed Python.

LD_LIBRARY_PATH

Run the following command to set the `$LD_LIBRARY_PATH` environment variable:

```
export LD_LIBRARY_PATH=$PYTHONHOME/lib
```

PATH

Run the following command to set the `$PATH` environment variable:

```
export PATH=$PYTHONHOME/bin:$PATH
```

Verify the Python installation

Run the following command to change directories to `$PYTHONHOME`:

```
cd $PYTHONHOME
```

Run the `ls` command to verify that `$PYTHONHOME` contains the following folders:

```
bin include lib share
```

Optionally, install third-party libraries

Optionally, use pip to install any Python third-party libraries in `$PYTHONHOME`.

For example, you can run the following commands to install numpy and scikit-learn:

```
bin/pip3 install numpy
bin/pip3 install scikit-learn
```

Copy Python to the Informatica domain

Run the following command to create an environment variable called `$INFA_HOME` and set the value to the location of the Informatica domain:

```
export INFA_HOME=~/Informatica/10.4.1.1/RC
```

Update the command based on the product version that you use.

Run the following command to copy the Python installation from `$PYTHONHOME` to the Informatica domain:

```
cp -r $PYTHONHOME/* $INFA_HOME/services/shared/spark/python/
```

Run the following command to check the contents of the Informatica domain:

```
ll $INFA_HOME/services/shared/spark/python/
```

Verify that the contents include the following folders:

```
bin
include
lib
share
```

Optionally, verify third-party library installations

Run the following command to change directories to the Informatica domain:

```
cd $INFA_HOME/services/shared/spark/python
```

Run the following command to invoke the Python interpreter:

```
bin/python3
```

Run the following Python code:

```
import numpy
import scikit-learn
```

Verify that the import statement is successful for each Python third-party library that you installed.

Press `Ctrl+D` to exit the Python interpreter.

Integrate Python with the Hadoop connection

To integrate the Python installation with the Hadoop connection, configure the Spark advanced properties that the Spark engine requires to run the Python transformation.

In the Administrator tool, complete the following tasks:

1. Navigate to **Manage > Connections**.
2. Select the Hadoop connection that connects to the Hadoop cluster.
3. Edit the **Spark configuration**.
4. Edit the **Advanced Properties**.
5. Configure the following advanced properties:
   infaspark.pythontx.exec

   Location of the Python executable binary. Set the property to the following value:

   `$INFA_HOME/services/shared/spark/python/bin/python3`

   infaspark.pythontx.executorEnv.PYTHONHOME

   Location of the Python installation directory. Keep the following default value:

   `$INFA_HOME/services/shared/spark/python/`

Install Jep for streaming mappings

Streaming mappings require Jep as part of the Python installation. If you run the Python transformation in a streaming mapping, complete the following tasks to install Jep:

Set JAVA_HOME

Jep has a dependency on JDK. Before you install Jep, you must set the environment variable `$JAVA_HOME`. The Informatica domain contains an installation of Java, so you can use the JDK that is packaged in the Informatica domain.

Run the following command to set `$JAVA_HOME` to the packaged JDK:

```
export JAVA_HOME=$INFA_HOME/java
```

Install Jep

Run the following command to change directories to `$PYTHONHOME`:

```
cd $PYTHONHOME
```

Run the following command to install Jep:

```
bin/pip3 install jep
```

Copy Python to the Informatica domain

Run the following command to remove all files from the following directory in the Informatica domain:

```
rm -rf $INFA_HOME/services/shared/spark/python/*
```

Run the following command to copy the Python installation from `$PYTHONHOME` to the Informatica domain:

```
cp -r $PYTHONHOME/* $INFA_HOME/services/shared/spark/python/
```

Integrate Jep with the Hadoop connection

To integrate Jep with the Hadoop connection, complete the following tasks in the Administrator tool:

1. Navigate to **Manage > Connections**.
2. Select the Hadoop connection that connects to the Hadoop cluster.
3. Edit the **Spark configuration**.
4. Edit the **Advanced Properties**.
5. Configure the following advanced properties:
   infaspark.pythontx.executorEnv.LD_PRELOAD

   Location of the Python shared library. For example, set the property to the following value:

   `$INFA_HOME/services/shared/spark/python/lib/libpython3.6m.so`

   infaspark.pythontx.executorEnv.PYTHONHOME

   Location of the Python installation directory. Keep the following default value:

   `$INFA_HOME/services/shared/spark/python/`

   infaspark.pythontx.submit.lib.JEP_HOME

   Location of the Jep package. For example, set the property to the following value:

   `$INFA_HOME/services/shared/spark/python/lib/python3.6/site-packages/jep/`

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